AgNW Reservoir Computing

Can the stochastic silver-nanowire (AgNW) network simulated by `AgNW_FDT_Memristor_Framework` work as a physical reservoir computer?

Answer

In short: yes in simulation, conditionally for real devices. A volatile-junction network read through 16 pad electrodes is a reproducible, modest reservoir (memory of about 4 inputs); whether real AgNW networks behave the same depends on their junction noise: random filament ruptures above about 10^{-4} per second per junction break it. Pinning that rate is the decisive step. N10-C (`docs/RESULTS.md` §21) maps the measured jump statistics of Milano et al. 2025 onto it; the estimate, 0.005-0.03 per second, is 50-300 times above the tolerance. No remedy tried so far restores the result at that noise level: a consolidated backbone, online retraining of the readout, a bias + small input operating point (the best noise-free reservoir, memory capacity 5.8, but the most fragile) and a junction voltage scale matched to the measured network. **Network size does work, mostly**: at 10^{-2} ruptures per second per junction, inside the real-junction range, 9 of 10 simulated 2.83 mm networks compute beyond their input (8 of 10 at 2 mm); the pooled criterion over all 10 fails on the NARMA-10 margin, and 4 mm passes it (3 networks only) (N10-L, N10-LS). Realistic noise still costs about half the noise-free capacity at every size; what size buys is reliability across networks. Measured devices (about 1 cm) are larger still (reports: `docs/report_2026-10-09/report.md` on 10 networks per size, `docs/report_2026-10-07/report.md` on network size, `docs/report_2026-10-06/report.md` on the noise question, `docs/report_2026-09-29/report.md` on the main result).

With the framework's junction model: no. With a volatile junction built from measured AgNW behaviour (N10): yes, weakly. The framework's binary junctions (Lamas 2026 preset) and a graded filament junction calibrated on Lamas Fig. 4 (N4) never compute more than the current input. Junctions that relax spontaneously on a time τ_s matched to the input rate (N10) make the network a weak but reproducible reservoir, the first readout to pass criterion G1.

1. Methodology

Everything below is the same for every model, so results are comparable. Design: `docs/PLAN.md`. Full results: `docs/RESULTS.md`.

1.1 Fixed network (imported, never copied)

Geometry, wires, and Kirchhoff solver come unchanged from `agnw_sim` (Lamas 2026 preset, framework commit `f6b6951`):

- $N = 2000$ wires, length 70 μm, on 1 mm^2; wire diameter 90 nm.
- Dual-node circuit with wire resistance; solved with the framework's cached solver (multi-electrode layouts use our own multi-terminal solve, tested to match the framework matrix to round-off).
- Integration step 1 ms (framework binary) or 2 ms (N4 filament, N10 volatile; both validated against 1 ms).
- 3 network seeds $\times$ 2 noise replicates per setting; all numbers are test scores at a validation-chosen operating point, mean $\pm$ sd with a 2-sd margin (see §1.5).

1.2 Interface: drive, electrodes, readout

- **Drive.** Scalar input $u_k \sim U(0,1)$ mapped to a piecewise-constant electrode voltage $V_k = V_{\mathrm{lo}} + (V_{\mathrm{hi}} - V_{\mathrm{lo}})u_k$, held for `hold` steps. Swept: hold time (10 ms to 3 s per input) and window position/width relative to the conduction onset. Multi-electrode encodings: same signal everywhere ("direct"), fixed random weights per electrode ("weights", $V_e(t) = V_c + \frac{1}{2}w_e(2u(t) - 1)$), or delayed copies $u(t - d_e)$ across electrodes.
- **Electrodes.** (a) Two bands (framework layout: left 5% driven, right 5% grounded). (b) Split bands (8 + 8 segments). (c) 4×4 array of 50 μm pads (8 driven + 8 grounded), plus a `pads_float` variant (4 driven + 4 grounded + 8 floating read pads). Windows are re-centred per layout from a quasi-static onset sweep (bands ≈ 2.1 V, pads ≈ 0.35 V).
- **Readouts.** Total `current` sampled densely within each hold (10--50 virtual nodes, so the transient is seen); `electrodes` (all 16 electrode currents, plus floating-pad voltages where present); 32 internal voltage `probes` (need micro-contacts; reference only); all `junctions` states (oracle upper bound, not measurable). Features below the 10^{-5} resolution floor (10 μV / 10 nA) are dropped.
- **Readout training.** Ridge regression (SVD, unpenalized intercept, features z-scored) on the first 60% of the stream; ridge α and operating point (hold, window) chosen on the next 20% (validation); all reported scores on the last 20% (test). Washout $\geq 3\times$ the slowest time constant precedes scoring, plus a drift check on the ON fraction.

1.3 Tasks, baselines, criteria

- **Tasks.** Linear memory capacity MC (sum of $r^2(u_{t-k})$, $k = 1..30$, only lags above the noise floor); degree-2 capacity (Legendre $P_2(u_{t-k})$ and products $u_{t-k}u_{t-l}$); NARMA-10 NMSE; sine/square per-sample accuracy; consistency (mean correlation across the two

replicates). Consistency is measured on the *state part* (I/V_{in}, V_p/V_{in}) and on its sample-to-sample *increments*, so shared drift and the V_{in} echo do not count.

- **Baselines (no nanowires).** Input only, linear delay line (32 taps), quadratic delay line, ESNs (16/32/200 units), Gaussian-noise floor. Same streams, same splits.
- **Go/no-go criteria** (fixed 2026-09-27 before the decisive runs, `agnw_rc/criteria.py`): **G0** reproducible (increment consistency ≥ 0.8); **G1** computes beyond the input (MC ≥ 1 with ≥ 2 lags above floor, **and** degree-2 ≥ 1, **and** NARMA-10 ≤ 0.70 vs input-only ≈ 0.80--0.84); **G2** useful (MC + degree-2 $\geq 50\%$ of an ESN of the same size, **and** sine/square ≥ 0.95). For multi-electrode readouts: **strict** (headline; ESN as large as the feature count incl. virtual nodes) and **loose** (one ESN unit per physical electrode; ESN-16 total 15.4, so ≥ 7.7).

2. Models explored: what failed, what worked

The wires never change. What changes is the **junction** (plus interface where noted). Each row below is self-contained: code, what the junction physically is, what interface it ran on, and the verdict.

Nomenclature. E = experiment planned upfront in `docs/PLAN.md` (E1 main sweep, E2 relaxation-rate check, E3 slow inputs, E4 decisive time-scale-matched test). **N** = follow-up work item from the diagnosis in `docs/NEXT_STEPS.md` (N1 mean-field ablation, N2 heterogeneous forgetting, N3 masks, N4 filament junction, N5 multi-electrode, N7 noise averaging, N8 trade-off map, N9 virtual nodes, N10 volatile measured-data junction). Suffixes refine an item: M1 = N1 runs, N10-1 = first N10 campaign (two bands, 32 probes), N10-E = N10 on 16 electrodes, N10-M = memory/noise limits, N10-S = sensitivity, N10-R = ruptures, N10-C = calibration on open data (deferred).

Colour code used in this section and in §3. green = worked / positive result (passes G1 or robust). **red** = failed / negative result (at or near input-only level, or breaks G1). **blue** = to be tested in the future. **violet** = parameter still to be pinned by measurement or literature (e.g. relaxation time, voltage scale, rupture rate).

model (self-contained)	idea tested	verdict
Framework binary stochastic (Lamas 2026 preset; runs E1--E3; two bands, total current / 32 probes / all junctions)	Reference junction: binary ON/OFF, voltage-driven SET, current-driven RESET, one relaxation rate $p_{\text{relax}} = 10^{-4}$ (10 s).	**fails.** MC ≤ 0.26 (0.76 at 1 s holds), degree-2 ≤ 0.13, NARMA ≥ 0.73, chance accuracy. Junction states irreproducible (consistency 0.00); only the global

model (self-contained)	idea tested	verdict
		conductance partly responds.
N1 mean-field (runs M1; same grids as E1, own windows 3.5--5.5 V)	Ablation, not a device: each junction carries its ON probability with the exact framework rates, no coin flips. Asks whether noise is the blocker.	**diagnostic, partial.** Reproducible (conductance consistency 1.00), nonlinear (degree-2 2.2, sine/square 1.00), but MC only 1.8 vs ESN-32 13.8. Noise blocks nonlinearity; the single slow rate still blocks memory.
N2 heterogeneous forgetting (mean-field + per-junction p_{relax} spread)	Same as N1 but each junction gets its own forgetting rate (log-uniform over 1--3 decades). Tests whether a spread of time scales fixes memory.	**helps $\sim$2x, then capped.** MC 1.8 $\rightarrow$ 2.6, degree-2 1.9 $\rightarrow$ 4.3. An ideal leaky-bank reference shows purely relaxational single-input memory is noise-limited (≈ 9 noise-free, 2--3 at realistic SNR).
N7a replicate averaging / N7b voltage confound (framework binary, one E1 point / M1 windows)	N7a: average $R = 1$--8 identical devices to wash out noise. N7b: rerun stochastic model at the high mean-field voltages to rule out a voltage confound.	**fails.** Junction/probe consistency stays ≤ 0.015 ($\sim$100+ devices needed); high voltage makes it worse. N1 gains come from removing noise, not from voltage.
N5a multi-electrode, random weights (binary junction) (8 + 8 split	Same binary junctions, richer interface: 8 inputs with fixed random weights, 8 grounded	**fails on the binary junction.** All scores at input-only level (MC ≈ 0, chance

model (self-contained)	idea tested	verdict
bands / 4×4 pads, 16 electrode currents)	outputs. Tests the single-electrode-pair bottleneck.	accuracy). A richer interface cannot extract what a noisy state does not hold.
N4 continuous filament, Fig. 4-calibrated (runs N4-1; two bands, 10--100 ms holds)	Physical graded junction: filament Λ grows above $V_{\text{set}} = 0.06$ V, dissolves below $V_{\text{reset}} = 0.03$ V over $\tau_d \approx 25$ s, Simmons tunnelling conductance; fitted to Lamas Fig. 4 I--V loop.	**fails (frozen).** Reproducible (junction consistency 0.69--0.83) but static: MC 0, degree-2 ≤ 0.02. Filaments saturate in $\sim$0.1 s or persist 30 s; most junction voltages sit in the hold band where nothing happens.
E4 time-scale-matched inputs (decisive, pre-registered G0--G2) (binary + calibrated N4; holds 0.3--3 s, washout $3 \times \tau$, 50 virtual nodes)	Same two junctions, inputs slowed to the network's own forgetting time. First run with validation-chosen operating point and 2-sd margins.	**fails G1 for both.** Binary current at 3 s: MC 0.94, degree-2 0.50, NARMA 0.71 (G0 yes, G1 no). N4 frozen at every hold (0 capacity, conductance varies $\leq 0.4\%$).
N8 short: hold band vs forgetting (N4 scan: $\tau_d = 0.3$--25 s $\times$ band ratio $r = V_{\text{reset}}/V_{\text{set}} = 0.5/0.9$; 1 net, 400 samples)	Disentangles why N4 freezes: is it slow forgetting or the wide hold band? Narrow band $r = 0.9$ keeps a Fig. 4 fit (loss 0.08) while unfreezing.	**diagnostic: band is the cause.** Wide band frozen at every τ_d ($\sim$20/5953 junctions move); narrow band moves 700--1700/sample, MC ≈ 1, degree-2 ≈ 1. Still far from G2. Full E4-style check of narrow-band N4 (N4-nb) still to be run.

model (self-contained)	idea tested	verdict
N10 volatile junction from measured data (runs N10-1; two bands, 32 probes + current)	Junction built from measured AgNW behaviour (Milano 2022 rate law, non-polar, relaxes spontaneously at zero bias with $\tau_s = 0.03$--3 s, spread across junctions, no hold band; power-gated consolidation + rupture). Settings $v_{\mathrm{scale}} = 5$, $k_{\max} = 100$ s^{-1} set onset ≈ 2.2 V and switching ≥ 10 ms. Best hold $\approx \tau_s/3$.	**works weakly: first G1 pass.** Probes at $\tau_s = 0.3$--3 s: MC ≈ 2.0--2.4, degree-2 ≈ 3.4--4.1, NARMA ≈ 0.48--0.51, sine/square 1.00, consistency 1.00 over 3 nets. G2 no (6 vs 14). Current-only readout just below G1.
N10-E: N10 on 16 pads, electrode currents (chip-compatible) (4×4 pads, 8 driven with random weights + 8 grounded; 100 ms inputs; 3 nets)	Same volatile N10 junction ($\tau_s = 0.3$ s), but 16 spatially distinct electrodes replace the internal probes; readout = 16 electrode currents with virtual nodes.	**works: G1 + G2 loose.** MC 3.48, degree-2 6.74, NARMA-10 0.40 ($\approx$ ESN-16 0.37), sine/square 1.00, consistency 1.00. 3x memory and 4x nonlinearity of the two-terminal current. G2 strict no (10.2 vs 25). Floating read pads add nothing.
N10-M: memory and noise limits on N10-E (same pads; slower τ_s, spread, delayed inputs, Gaussian filament noise σ)	Stresses the working N10-E setup: slower junctions with matched holds, wider τ_s spread, delayed input copies across electrodes, additive filament noise.	**limits mapped (mixed).** Own memory capped ≈ 4 inputs for any τ_s; $\approx 1\%$-range noise tolerated (G1 holds). $\approx 4\%$ noise breaks G0/G1 (consistency 0.47); delays add only interface memory (MC 7.6 = delay line 7 + 0.6).

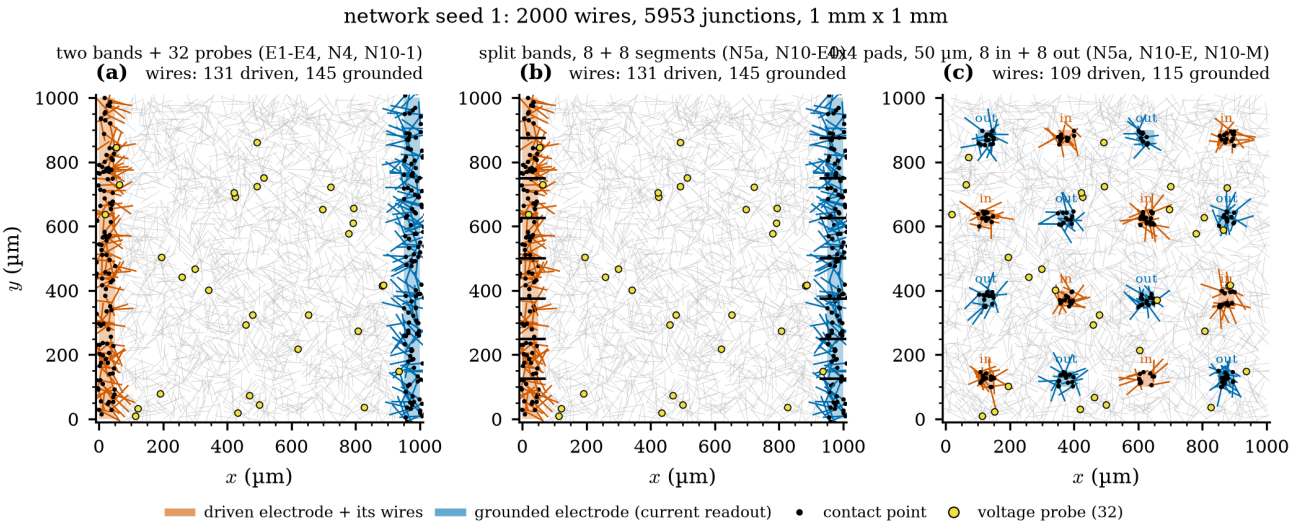
model (self-contained)	idea tested	verdict
N10-S: sensitivity to chosen settings (N10-E setup; voltage scale 3--8, switching cap 30--300 s $^{-1}$)	Repeats N10-E while varying the two hand-chosen N10 settings. Each setting gets its own onset sweep.	**robust.** All scores within 1 sd of N10-E, all G0--G2 verdicts unchanged. Onset $\propto 1/v_{\mathrm{scale}}$. What still matters: τ_s (sets usable input rate) and junction noise.
N10-R: realistic rupture noise + mitigations (N10-E setup; random complete ruptures $r = 10^{-5}$--3 /s per junction; bigger pads, longer holds, double presentation)	Replaces smooth noise by abrupt Poisson filament ruptures (as in measured relaxation), and tries three mitigations at the breaking noise level.	**breaks above $\approx 10^{-4}$ /s; no mitigation restores G1.** 10^{-5} /s safe; up to 3×10^{-4} /s the typical run is unaffected, but single runs collapse from 10^{-4} /s (a rupture on a bottleneck junction invalidates the trained readout); from 10^{-3} /s every run loses about half its capacity. Larger pads worse. Online retraining and consolidation-on N10 still to be run.
N10-C: calibration on Milano 2025 measured data (their constant-voltage protocol on N10 two-band networks, 1--2.83 mm, sampled every 0.6 s like the instrument)	Turns their 0.08 detected conductance jumps per second (7 mm network) into a per-junction rupture rate: count the junctions whose rupture is detectable, extrapolate with network size.	**real junctions too noisy for a 1 mm reservoir.** Data reproduced (0.080 jumps/s, half up/half down). $r^* \approx 0.005$--0.03 /s per junction: 50--300x above the N10-R tolerance (extrapolated; rests on the largest

model (self-contained)	idea tested	verdict
		networks, few jumps). N10 junctions switch at $\sim 10\times$ their voltage (onset 14 V vs 2.4 V at 7 mm).
N10-RN: measured fluctuations on the readout (Milano 2025 network-level noise, OU + jumps, on the 16 electrode currents of N10-E; 1--30x measured; no new simulations)	Separates the size of the fluctuations the readout sees from what ruptures do inside the network.	**harmless at the measured level.** G0/G1/G2-loose hold at 1x; G1 up to 3x (per electrode) and 30x (common). The rupture damage in N10-R comes from rerouting inside a small network.
N10-K: consolidated backbone vs ruptures (N10 with the consolidated filament y on, 10 s forming drive; backbones 1.7--12.7 % of junctions; ruptures 0--10^{-2} /s)	Protects the bottleneck junctions with stable (thick) filaments, as in measured STP $\rightarrow$ LTP consolidation.	**no G1.** Collapses gone (1 of 72 runs; ruptures cost 4--25 %), but the backbone itself halves memory and cuts degree-2 by 60--70 %: robustness and computation compete for the same junctions. Input windows not yet re-tuned after forming.
N10-OR: online retraining of the readout (readout retrained every 10--40 inputs on the last 150--600, N10-R runs)	Lets the readout follow the network after a rupture reroutes the current.	**no help.** Never better than a static readout; short windows are too few samples for 800 features.
N10-B: bias + small input (bias 1.5--3x onset,	Operates deep in the conducting state, as the measured reservoir	**best noise-free reservoir so far**: MC

model (self-contained)	idea tested	verdict
input 3--15 % of the bias, as Milano 2025)	does.	5.8, degree-2 10.9, NARMA 0.36 (> ESN-16). Much more fragile: 32/36 runs collapse at 10^{-3} /s; measured-size readout noise alone breaks G1.
N10-V: voltage scale matched to Milano (`v_scale` 40)	Junctions switch at the measured network's voltage.	Reservoir unchanged, slightly more robust; calibration (r^*) unchanged; G1 still fails at $\geq 10^{-3}$ /s.
N10-L: larger networks (N10-E enlarged to 1.41, 2 and 2.83 mm, same density, pads scaled; ruptures 0--10^{-2} /s)	Tests whether the rupture tolerance grows with size (each rupture matters less in a bigger network).	**yes: 2.83 mm meets G1 at 10^{-2} /s** (MC 2.2, degree-2 3.4, NARMA 0.54), inside the real-junction range; 2 mm up to 3×10^{-3} /s. Size buys reliability (spread between networks shrinks ~3x), not capacity: about half the noise-free capacity is lost at every size.
N10-LS: 10 networks per size (networks 4-10 at 1, 2, 2.83 mm, 0 and 10^{-2} /s; ruptures + junction + readout noise at 2.83 mm; 4 mm, 3 networks; HPC cluster, 348 runs)	Checks the N10-L headline with publication-level statistics; decision rules fixed in advance (docs/PLAN_N10L_SEEDS.md).	**9/10 networks pass G1 at 2.83 mm, 8/10 at 2 mm; pooled G1 fails (NARMA margin)**. Size halves the spread between networks (ratio 1.98 [1.15, 4.20]); about half the capacity lost at every size; combined noise ~

model (self-contained)	idea tested	verdict
		ruptures alone; 4 mm passes pooled (3 networks).

Why the framework model fails: seen through its electrodes the network behaves as **one noisy, slowly relaxing integrator**. Which junctions switch is set by noise, so individual junction states and local probe voltages are not reproducible, and the network forgets over 10 s while inputs change every 10--100 ms. Why N10 works: volatile filaments with no hold band and τ_s matched to the input give many reproducible, diverse, fading variables; 16 spatially distinct electrodes expose them.



Layouts: two bands (E1--E4, N10-1; total current + 32 internal probes, left), split bands and 4×4 pads (N5a, N10-E/M/S/R; 16 electrode currents, middle/right). Orange driven, blue grounded, black contacts, yellow probes.

3. Headline numbers (same colour code)

800-sample streams, test data, operating point chosen on validation, means over 3 networks $\times$ 2 replicates. **Green rows** pass G1; **red rows** do not compute beyond the input.

best readout and setting	memory capacity	degree-2 capacity	NARMA-10 NMSE	sine/square accuracy
input only (no network)	0.0	0.0	0.84	0.50
framework binary junctions (E4, current, 3 s per input)	0.94	0.50	0.71	0.53

best readout and setting	memory capacity	degree-2 capacity	NARMA-10 NMSE	sine/square accuracy
Fig. 4-calibrated filament junctions (N4, E4)	0.0	0.0	0.84	0.50
volatile N10 junctions (N10-1: volatile Milano-law filament, no hold band, $\tau_s = 3$ s; 32 probes, 1 s per input)	**2.35**	**4.13**	**0.48**	**1.00**
volatile N10 junctions on 16 pads (N10-E: same junction on a 4×4 pad array, 16 electrode currents, 0.1 s per input)	**3.48**	**6.74**	**0.40**	**1.00**
ESN, 32 units (digital reference)	13.5	14.2	0.23	1.00
linear delay line, 32 taps (memory reference)	30.0	0.0	0.15	0.99

N10 meets G0 and G1; G2 only per physical contact (vs ESN-16), not per recorded number (strict).

How to read the figures. Each figure plots one score (MC, degree-2, NARMA error) vs the varied knob (hold time, τ_s, noise). Higher is better for capacities and accuracy, lower is better for NARMA NMSE. Red dotted lines = G1/G0 thresholds; magenta = G2 loose (dotted) / strict (dashed); grey = reference (two-band N10-1 or noise-free N10-E). Good = green-level points sitting above red with small error bars (3 nets); bad = red-level points at the input-only floor.

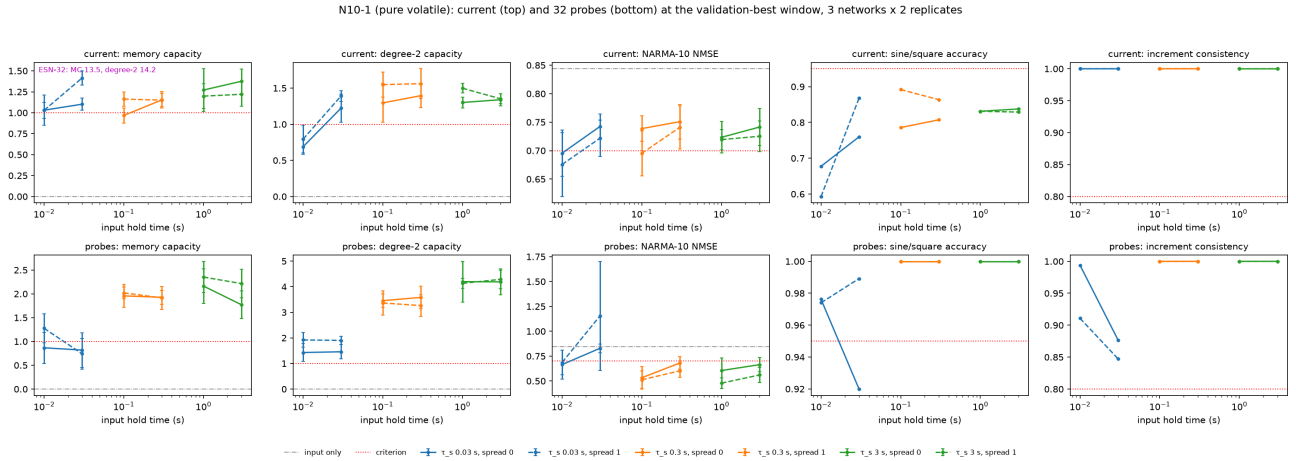


Fig. 15 (N10-1: volatile junction, two bands). Top: total current; bottom: 32 probes; x = hold time; solid/dashed = no spread / spread τ_s. Read: probes at 0.1--1 s holds clear all three red G1 lines with consistency 1.00 — the first green result; current stays red (just below G1). Longer 3 s holds do not help.

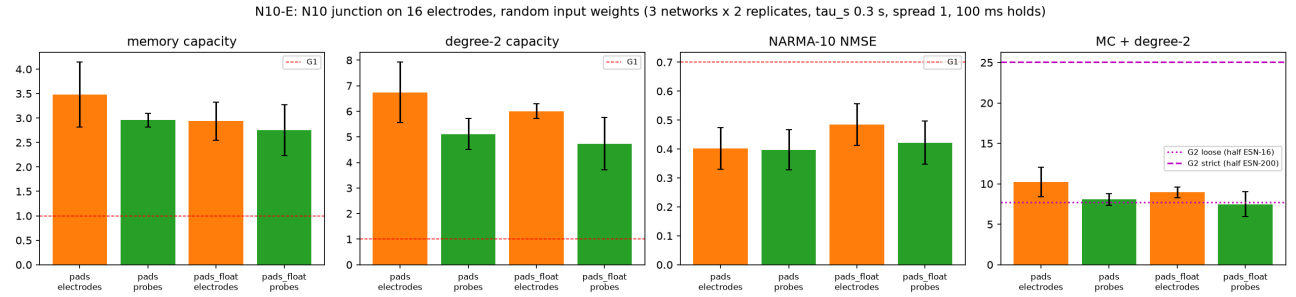


Fig. 17 (N10-E: same junction on 16 pads). Orange = 16 electrode currents (chip-measurable), green = 32 internal probes for reference. Read: orange clears G1 on all three panels and the magenta loose-G2 line, but not the strict-G2 line — green per contact, red per recorded number. Memory panel: 3.5 vs ESN-16 9.4 is the weak point.

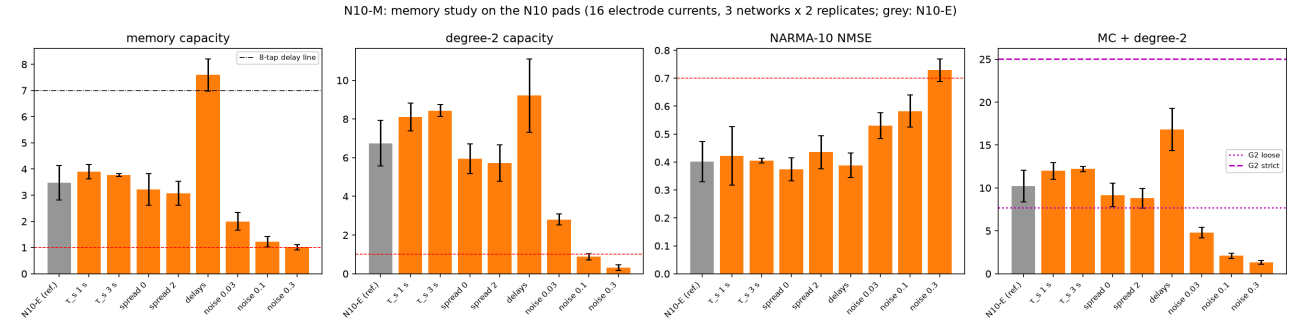


Fig. 18 (N10-M limits on the pad setup). Grey = N10-E reference. Read, left to right: slower τ_s barely moves memory (capped ≈ 4 — green but flat); delays lift MC to 7.6 but only to the delay-line level (interface memory, not network memory); noise $\sigma \geq 0.1$ ($\approx 4\%$ of filament range) drops all panels to red and consistency below G0.

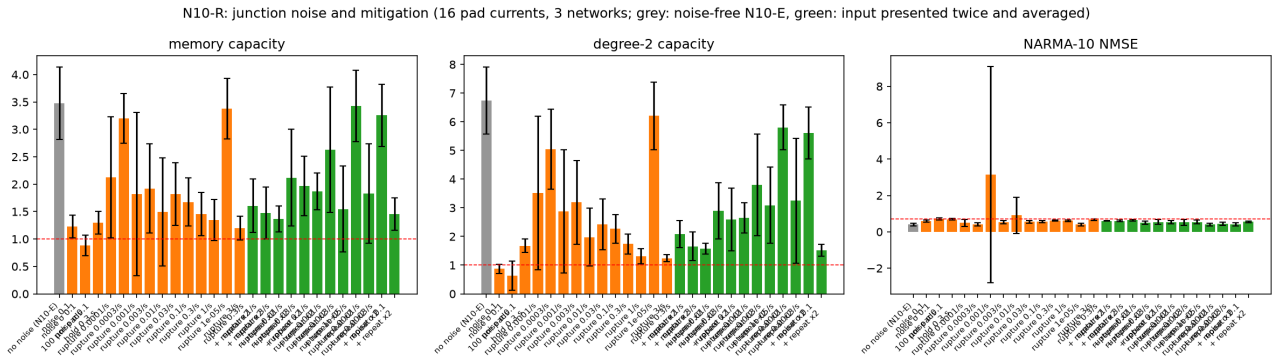


Fig. 20 (N10-R ruptures). Grey = noise-free N10-E; orange = rupture rates / mitigations; green = each input shown twice and averaged. Read: every orange point from 0.01 to 3 ruptures/s sits red (G1 broken, capacities halved); double presentation recovers only 10--40%. Lower rates (RESULTS §20): tolerance about 10^{-4} /s per junction; online retraining is still open.

A project review (2026-09-27) corrected consistency and memory-cap statements; see docs/REVIEW_2026-09-27.md. Work continues in docs/NEXT_STEPS.md: next steps are noise, consolidation and more electrodes on the N10 junction, and measured relaxation data to pin its parameters.

Full write-up with figures: **docs/RESULTS.md**. Design and definitions: docs/PLAN.md. Junction physics and framework audit: docs/SWITCHING_MECHANISMS.md. N10 design: `docs/N10_DESIGN.md`. Measurement request: `docs/MEASUREMENT_REQUEST.md`.

4. What is fixed and what this project adds

- **Fixed (from the framework, imported unchanged):** network geometry (N = 2000 wires, 70 µm, 1 mm^2), dual-node Kirchhoff circuit with 90 nm wires, stochastic SET/RESET dynamics with the Lamas 2026 parameters, exponential relaxation $p_{\text{relax}} = 10^{-4}$, and the cached solver. Simulations run through `agnw_sim.simulate.run_simulation`. `agnw_rc.config` holds the exact settings, and a test checks them against the framework preset.
- **Added here:**
  - input encoding: piecewise-constant electrode voltage, with a sweep over hold time and voltage window; random-weight and delayed multi-electrode encodings;
  - readout scenarios: two-terminal current with virtual nodes, 16 electrode currents (split bands / pads / pads with floating reads), 32 voltage probes, and all junction states as an oracle;
  - junction models: mean-field ablation (N1/N2), continuous filament calibrated on Fig. 4 (N4), volatile measured-data junction with consolidation, rupture and noise (N10 family);

- multi-terminal Kirchhoff solver and layouts (`electrodes.py`, `multicircuit.py`, `multidrive.py`);
- a ridge readout with validation-selected regularization and held-out test scores;
- tasks: memory capacity, degree-2 information-processing capacity, integration (EMA), NARMA-10, sine/square classification, and increment consistency;
- no-nanowire baselines: input only, delay lines, quadratic delay line, echo-state networks, and a noise floor;
- go/no-go criteria G0--G2 with validation-based operating point and drift check (`criteria.py`).

5. Layout

```
agnw_rc/            reservoir layer on top of agnw_sim
  config.py         framework working model + experiment grids (E1-E4, M1, N2, N5a, N4, N10-
1/E/M/S/R)
  drive.py          input -> run_simulation -> observables (current, probe voltages, junction
states)
  tasks.py          input streams and targets (memory, Legendre degree-2, EMA, NARMA-10,
sine/square)
  readout.py        SVD ridge regression, alpha chosen on a validation segment
  metrics.py        capacity, NMSE, accuracy, consistency (incl. increments)
  baselines.py      input only, delay lines, ESN, noise features
  evaluate.py       scoring of reservoir runs and baselines
  meanfield.py      noise-free (mean-field) junction dynamics + fast solver (N1/N2)
  electrodes.py     electrode layouts: two bands (framework), split bands, pad array (N5, N10-E)
  multicircuit.py   multi-terminal Kirchhoff solver on the framework's dual-node graph (N5, N10-
E)
  multidrive.py     multi-electrode drive/readout time loop, input encodings (N5, N10-E/M/R)
  filament.py       continuous-filament junction model (N4): rate law, Simmons tunnelling, noise
  fig4.py           Lamas Fig. 4 protocol and calibration loss for N4
  criteria.py       go/no-go criteria G0-G2, validation-based operating point, drift check
(review plan)
  volatile.py       N10 volatile two-time-scale junction (Milano rate law + consolidation,
ruptures, noise)
scripts/
  pilot.py, meanfield_pilot.py, multi_pilot.py   operating-point pilots -> results/*.npz
  run_sims.py       parallel, resumable simulations (e1 e2 e3 m1 n2 n7a n7b n5a n4_1 e4 n10_1
n10m;
                    --holds/--reverse/--limit split work across machines) -> results/raw/*.npz
(git-ignored)
  e4_report.py, n10_report.py, n10m_report.py, n10s_sensitivity.py, n10r_noise.py,
n10e_pilot.py, ...
                    go/no-go reports -> results/*criteria.json, figures/fig1*.png
  n10c_data.py, n10c_sim.py   N10-C: Milano 2025 open data (data/milano2025/) and their protocol
on N10
  n10e_readout_noise.py, n10k_consolidation.py   N10-RN measured noise on the readout; N10-K
consolidated backbone
  n10r_retrain.py, n10b_bias.py, n10v_vscale.py   N10-OR online retraining; N10-B bias + small
input; N10-V matched v_scale
  n10l_size.py      N10-L larger networks (RunSpec.side, pads_scaled; --share splits runs across
machines)
  evaluate_all.py   all scores -> results/metrics.json, results/summary.csv
  n4_stride_check.py, n4_substep_check.py, n10_kmax_check.py, n10_stride_check.py, ...
                    integration-step validation
  n8_short.py       N4 hold-band vs forgetting scan -> results/n8_short.json
  effective_dimension.py, leaky_bank_reference.py, noise_averaging.py, switching_rates.py, ...
                    references and diagnostics
  make_figures.py   all figures (--only, --mode paper|presentation) -> figures/*.pdf|png
figures/README.md   index: each figure -> drawing function, data, generating script
tests/              pytest (80 tests): framework contract, drive alignment, readout, metrics,
tasks, baselines,
                    mean-field, filament, volatile, multi-electrode, criteria
```

```
docs/PLAN.md        question, design, experiments, tests
docs/RESULTS.md     results, figures, interpretation, limitations
docs/NEXT_STEPS.md  diagnosis + ranked work items with status (start here when resuming)
docs/SWITCHING_MECHANISMS.md  junction ON<->OFF mechanisms, their rates at the operating points,
framework docs audit
docs/N4_DESIGN.md, docs/N10_DESIGN.md  junction models: equations, parameters, calibration and
experiment plan
docs/REVIEW_2026-09-27.md  full project review: bugs, corrected conclusions, evidence table
docs/LITERATURE_NOTES.md, docs/MEASUREMENT_REQUEST.md, docs/report_2026-09-29/,
docs/report_2026-10-06/  literature, experiment request, reports
memory/             portable copy of the project memory (loaded via CLAUDE.md)
legacy/             earlier prototype, kept for reference only (its memory-capacity numbers are
invalid)
```

6. Setup

The framework is installed in editable mode from its own checkout. Nothing is copied from it.

```
FRAMEWORK=../AgNW_FDT_Memristor_Framework     # sibling checkout; adjust if it lives elsewhere
python3 -m venv .venv
.venv/bin/pip install -e "$FRAMEWORK"
.venv/bin/pip install -e ".[dev]"
.venv/bin/python -m pytest            # 80 tests, ~7 min
```

7. Reproduce

```
.venv/bin/python scripts/pilot.py
.venv/bin/python scripts/run_sims.py e1 --procs 14
.venv/bin/python scripts/run_sims.py e2 --hold 100 --v-center 1.75
.venv/bin/python scripts/run_sims.py e3 --v-center 1.75
.venv/bin/python scripts/meanfield_pilot.py
.venv/bin/python scripts/run_sims.py m1 --procs 14     # N1 mean-field ablation
.venv/bin/python scripts/run_sims.py n2 --procs 14     # N2 heterogeneous forgetting
.venv/bin/python scripts/evaluate_all.py --procs 14
.venv/bin/python scripts/run_sims.py e4 --model stochastic --procs 12  # E4 binary, ~6 h
.venv/bin/python scripts/run_sims.py e4 --model filament --procs 16    # E4 N4 at 2 ms, ~5 h
.venv/bin/python scripts/e4_report.py
.venv/bin/python scripts/run_sims.py n10_1 --procs 16  # N10-1, 216 runs, ~12 h
.venv/bin/python scripts/n10_report.py
.venv/bin/python scripts/n10e_pilot.py sweep; .venv/bin/python scripts/n10e_pilot.py run;
        .venv/bin/python scripts/n10e_pilot.py report
.venv/bin/python scripts/n10e_pilot.py full --procs 18; .venv/bin/python scripts/n10e_pilot.py
        full_report
.venv/bin/python scripts/run_sims.py n10m --procs 18; .venv/bin/python scripts/n10m_report.py
.venv/bin/python scripts/make_figures.py            # --mode presentation for talk-sized fonts
```

Each script sets one BLAS thread per process, because the scripts already run in parallel over processes. Leaving this out oversubscribes the CPU badly. The raw runs (~6 GB, without junction states; full files archived on an external drive, see RESUME_FROM_HERE.md) are git-ignored. `results/metrics.json` and `results/*criteria.json` keep every score. Full command list: `docs/RESULTS.md` §28.

8. Next steps

Tracked in **docs/NEXT_STEPS.md**: the diagnosis (stochastic binary switching, a single forgetting time scale, a single electrode pair), work items ranked by gain/effort, their status, and a log. Decisive unknown: the per-junction rupture rate (tolerance about 10^{-4} /s). N10-C calibration on Milano 2025 open data gives 0.005-0.03 /s, far above it (Gaussian part $\sigma \lesssim 0.05$, at the edge). Remedies tried and failed: consolidated backbone, online retraining, bias + small input, matched voltage scale (RESULTS 23-26). Next: larger reservoir networks; single-junction / small-network measurements per `docs/MEASUREMENT_REQUEST.md` would settle it.